

TECHNICAL QUALITY REPORT

Roadwatch Vision Function Audit

What works, what was fixed, what was removed, and what should change next

16 / 16	0	2	2
Automated tests passing	Known failing tests	Duplicates removed	Files changed

Executive summary

The audited Python application compiles successfully and its full available automated test suite passes. One confirmed cloud-playback defect was repaired: the newer playback layer expected Supabase private-bucket and signed-URL support that the underlying service did not yet provide. Two identical helper implementations were also consolidated into shared utilities.

No tested function is currently known to be broken. The principal remaining risk is coverage: hardware capture, OCR, encoding, external notification delivery, and live cloud integrations cannot be fully proven by the existing unit tests.

Audit basis

Check	Result	Meaning
Python compilation	PASS	Application, core, service, and test modules compile.
Automated tests	PASS - 16 tests	Authentication, API, storage, GPS, video metadata, cloud playback, and dual-camera AI behaviors passed.
Change validation	PASS	Updated files compile and the complete tests still pass.
Hardware/external systems	PARTIAL	Real cameras, codecs, OCR engines, SMTP/webhooks, Firebase, and Supabase require environment-specific integration tests.

1. Functions and capabilities confirmed working

Area	Confirmed behavior	Evidence
Authentication	Password hashing and verification; JWT issue/decode round-trip; admin authentication.	Automated tests
API security	JWT required on protected endpoints; role checks enforced; sync endpoint authorization.	Automated tests
Video access safety	Video IDs cannot escape the configured metadata directory.	Automated tests
GPS	Browser GPS rejects invalid latitude and longitude values.	Automated tests
Local JSON storage	Missing data returns an independent default value rather than shared mutable state.	Automated tests
Settings	Credential-like values are not persisted into the local settings file.	Automated tests
Storage cleanup	Original and processed evidence are excluded from cache-removal candidates.	Automated tests
Export cleanup	Only failed temporary export artifacts are removed.	Automated tests
Upload validation	Cloud upload accepts only an existing, compressed MP4 playback artifact.	Automated tests
Video metadata	Repair logic restores MP4 playback fields from available metadata.	Automated tests
Cloud normalization	Stable Supabase identifiers, MIME type, byte size, and cloud readiness are normalized.	Automated tests
Cloud playback	Valid cached HTTPS URLs are reused; expired URLs are refreshed; local paths are rejected.	Automated tests
Private Supabase playback	Signed-URL endpoint is used for private buckets.	Automated tests
Dual-camera AI	AI frames are passed through the configured vision pipeline.	Automated tests

Statically healthy modules

The following modules compile cleanly and remain available to the application. Compilation confirms syntax and import structure, but should not be mistaken for complete runtime coverage.

Module group	Functions and responsibilities
core.annotation / detector / tracker / movement / ocr	Frame annotation, YOLO loading/detection, object tracking, movement detection, and OCR scanning.
core.video_io / vision_pipeline	Writer selection, video conversion/compression, playback selection, cleanup, and AI pipeline orchestration.
core.dual_camera / opencv_recorder	Camera configuration, capture, previews, recording, AI sampling, health, session summaries, and camera probing.
services.video_* / compression / detection_log	Recording metadata, processing, compression, thumbnails, and detection persistence.
services.incident / stolen_vehicle / vehicle_profile / contact	Incident lifecycle, stolen-vehicle reports, vehicle profile, and emergency contacts.
services.firebase / supabase / cloud_playback	Cloud metadata, storage upload/listing, playback URL creation, and normalization.
services.gps / notification / sync / upload_queue	Location tracking, notifications, remote sync, and upload task state.
api_app / streamlit_app	REST endpoints, authorization dependencies, pages, dashboards, playback, and administrative UI.

2. Changes completed during the audit

Change	Previous condition	Completed result
Supabase constructor contract	Cloud playback passed bucket privacy and expiry arguments that SupabaseService did not accept.	Constructor now supports bucket_public and signed_url_expiry_seconds while preserving older callers.
Signed playback URLs	SupabaseService had no create_playback_url method.	Public buckets return a public URL; private buckets POST to the signed-object endpoint with a bounded TTL.
Playback validation	No service-level validation for empty object paths or invalid expiry values.	Missing paths fail clearly; expiry input is normalized to a positive integer.
Refresh behavior	refresh_video_url always returned a public URL.	Refresh now uses the correct public or signed playback path and includes timestamps.
Duplicate format_duration	A helper was repeated in streamlit_app.py and services/report_service.py.	Streamlit imports the shared report utility; duplicate removed.
Duplicate is_remote_video_source	A helper was repeated in streamlit_app.py and core/video_io.py.	Streamlit imports the shared video utility; duplicate removed.

Files changed

File	Purpose
services/supabase_service.py	Private/public playback support, signed URL generation, expiry metadata, and validation.
streamlit_app.py	Removed duplicate helpers and imported canonical implementations.

3. Changes still recommended

Priority	Change	Reason / acceptance condition
P1	Add tests for camera and video failure paths	Mock camera-open failure, writer/codec failure, dropped frames, thread shutdown, partial files, and processing exceptions.
P1	Add cloud integration tests	Verify upload, object listing, signed URL expiry, Firebase write/read, pagination, and retry behavior against non-production test projects.
P1	Expose cloud privacy settings consistently	Add bucket-public and signed-URL-expiry settings to the application configuration and pass them through every SupabaseService construction site.
P2	Split streamlit_app.py	The file exceeds 3,000 lines. Move pages, playback helpers, settings, and admin flows into focused modules without changing behavior.
P2	Add pytest, coverage, and Ruff	Standardize test execution, measure untested branches, and catch unused imports, unreachable code, and style defects automatically.
P2	Test notifications and GPS background service	Cover SMTP/webhook failure, timeouts, stop events, device errors, reverse-geocoding failure, and status transitions.
P2	Resolve the Starlette/httpx deprecation	Update compatible dependency versions so API tests run without the TestClient deprecation warning.
P3	Document compatibility wrappers	Mark intentionally retained aliases such as old writer entry points and define when they may be removed.

4. Functions requiring repair or removal

Known broken functions

None remain among the behaviors covered by the current automated tests. The previously failing private-bucket playback path was repaired and its test now passes.

Removed as redundant

Removed implementation	Canonical implementation retained	Why removal was safe
streamlit_app.format_duration	services.report_service.format_duration	Identical formatting behavior; only the import location changed.
streamlit_app.is_remote_video_source	core.video_io.is_remote_video_source	Identical HTTP/HTTPS check; all Streamlit call sites use the imported helper.

Intentionally retained

Function family	Decision
Video writer helpers	Retained because general, MP4-specific, WebM-compatibility, and dual-camera writers have different return contracts and failure behavior.
Firebase convenience wrappers	Retained as a module-level public API over FirebaseVideoService.
Cloud playback convenience wrappers	Retained for environment-driven callers that do not construct service objects directly.
utc_now helpers	Retained within separate service modules to avoid unnecessary coupling; they are small and context-local.

5. Verification and limitations

Verification	Outcome
unittest discovery	16 tests executed; all passed.
compileall	Application Python sources compiled without syntax failures.
Change regression check	Full test suite remained green after both fixes and consolidation.
Known warning	FastAPI TestClient emits a Starlette/httpx compatibility deprecation warning.
Tooling gap	pytest, coverage, and pyflakes/Ruff are not installed in the project virtual environment.

Scope note

A passing unit test proves the tested behavior under its controlled inputs. It does not prove physical camera availability, local codec support, OCR model accuracy, network connectivity, cloud permissions, SMTP delivery, or webhook availability. Those items should be validated in the deployment environment using the recommended integration tests.

Recommended next milestone

Implement the P1 test and configuration items first. The target exit criteria should be: repeatable camera failure simulations, non-production cloud round-trips, and one consistent bucket-privacy configuration used by every playback and upload path.